

Pongal



Rice kheer

Fig 10.5: Complementary proteins

ACTIVITY - 1

Match the right answer

1) Polypeptides	-	Essential amino acid
2) Legumes	-	Complete protein
3) Proline	-	Conjugated protein
4) Arginine	-	Non-essential amino acid
5) Tryptophan	-	Derived protein
6) Meat	-	Simple protein
7) Nucleoproteins	-	Semi-essential amino acid
8) Albumin	-	Incomplete protein

Complementary proteins : If two sources of incomplete proteins are combined in the same meal, the resulting protein may be of better quality. These are called as ***Complementary proteins*** (e.g) Pongal prepared using moong dhal and rice is of better quality than rice or dhal cooked separately. Rice is deficient in aminoacid lysine, but rich in methionine. Pulses are rich in lysine, but deficient in methionine. So, rice and pulse combination will complement each other. Rice Kheer is another example, where animal and vegetable proteins –milk and rice are cooked together.

10.4 Food sources of proteins

Animal sources are complete proteins which includes meat, egg, fish and poultry and they are good protein foods in both quantity and quality. Milk is a valuable source of protein (casein) because although it does not contain a large quantity of protein, the quality is excellent.

Good sources of plant proteins are legumes, pulses, nuts and oil seeds, but their quality is poorer than that of animal foods. However, complementing two plant sources or combining an animal and a vegetable source in one meal increases the nutritional value of the meal tremendously. All vegetables and fruits are poor sources of proteins.

ACTIVITY - 2

- Display food sources of complete and incomplete proteins
- Prepare and display any dish which is an example of complementary proteins

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